VOX for Beads — User Manual

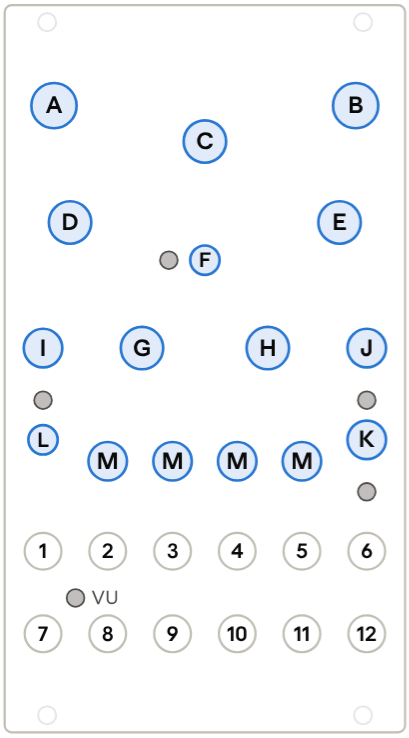
Oct 14, 2026 · @graeme neilson

VOX is alternative firmware for Mutable Instruments Beads that turns the module into a processor for spoken word: speeches, interviews, adverts, field recordings of voices. It understands what it is listening to: where the syllables and words are, which sounds are vowels and which are consonants, the pitch of the voice and the shape of its vocal tract. The vox_multi firmware holds six algorithms built on that analysis; hold MEDIA at power-on to choose one. Each algorithm also exists as a single firmware.

Front panel

VOX uses the Beads panel unchanged. Throughout this manual, knobs and buttons are referred to by a letter in square brackets, such as TIME **[D]**, and jacks by a number in parentheses, such as IN L **(7)**. The drawing is to scale.

Front panel key: controls [A]–[M], jacks (1)–(12)



Knobs and buttons
[A] FREEZE button
[B] SEED button
[C] DENSITY
[D] TIME
[E] PITCH
[F] MEDIA button, with its RGB LED to the left
[G] SIZE
[H] SHAPE
[I] FEEDBACK
[J] DRY/WET
[K] REVERB
[L] ASSIGN button
[M] Attenuverters for TIME, SIZE, SHAPE, PITCH

Jacks
(1) ASSIGN CV in
(2)–(5) TIME, SIZE, SHAPE, PITCH CV in
(6) DENSITY CV in
(7) IN L: the voice
(8) IN R: second input
(9) FREEZE gate in
(10) SEED gate or clock in
(11) OUT L
(12) OUT R, or the gate output

LEDs
ASSIGN LEDs: one under FEEDBACK, DRY/WET, REVERB
VU LED: between IN L and IN R

Beads front panel · the letters and numbers used in this manual

The four attenuverters **[M]** sit above the CV inputs they scale: TIME **(2)**, SIZE **(3)**, SHAPE **(4)** and PITCH **(5)**. The FREEZE and SEED buttons are lit from inside.

Installation and firmware update

VOX is installed with the Beads audio bootloader, exactly like an official firmware update. Nothing has to be opened, and the official firmware can be reinstalled the same way at any time.

File	Contents	Playing time
`vox/multi/prebuilt/vox_multi.wav`	All six algorithms, chosen at power-on	about 2 min 35 s
`vox/<name>/prebuilt/vox_<name>.wav`	One algorithm (psola, lpc, syllable, split, sequencer or chain)	about 1 min 45 s

Each `prebuilt` folder also holds the `.bin` file and a `SHA256SUMS` file with its checksum.

Installing

1. Power the module off.
2. Hold the FREEZE button **[A]** and power the module on. The bootloader starts.
3. Connect the output of a computer or phone to IN L **(7)** and play the `.wav` file at a moderate volume. Turn off any effect, equalizer, "sound check" or loudness normalization on the player. The TIME knob **[D]** sets the input gain of the bootloader.
4. When the transfer is complete, the module restarts into the new firmware.

If the transfer fails, power the module off and start again from step 1. The bootloader itself is never overwritten.

Going back to the official firmware

Install the official Beads `.wav` file with the same procedure.

What happens to the settings

- The CV calibration is shared with the official firmware and is never modified by VOX.
- VOX keeps the official settings layout and stores its own settings in bytes that the official firmware ignores. Swapping between VOX and the official firmware never makes either one reset the settings or start its factory test.
- The ASSIGN routing and amounts, the manual input gain and the gate output mode are shared by all firmwares.
- Each VOX algorithm remembers its own MEDIA sub-mode and options. In vox_multi this is true per algorithm, so switching between algorithms does not lose them.

Building from source

From the root of the eurorack repository, with an `arm-none-eabi-gcc` toolchain, and `python3` with `numpy` for the `.wav` file:

```
make -f vox/multi/makefile TOOLCHAIN_PATH=/path/to/toolchain/
make -f vox/multi/makefile TOOLCHAIN_PATH=/path/to/toolchain/ wav
```

Replace `multi` with `psola`, `lpc`, `syllable`, `split`, `sequencer` or `chain` to build a single firmware. `make -f vox/test/makefile check` runs the host test suite.

Selecting an algorithm

In vox_multi, hold MEDIA **[F]** while powering the module on, choose the algorithm with PITCH **[E]**, and press MEDIA again to start it. Without MEDIA held, the module starts directly with the algorithm used last.

1. Power the module off.
2. Hold MEDIA **[F]** and power the module on, then release MEDIA. The FREEZE **[A]** and SEED **[B]** LEDs blink alternately: the module is waiting for a choice and makes no sound.
3. Turn PITCH **[E]**. Its travel is divided into six equal zones, from algorithm 1 fully counter-clockwise to algorithm 6 fully clockwise. The MEDIA LED and the three ASSIGN LEDs show the current choice.
4. Press MEDIA **[F]** to confirm. The algorithm starts, and the choice is remembered for the next power-on.

Zone (PITCH)	Algorithm	MEDIA LED	ASSIGN LEDs lit (FEEDBACK · DRY/WET · REVERB)
1 (fully CCW)	PSOLA: pitch-synchronous voice	green	● ○ ○
2	LPC: analysis and resynthesis	red	○ ● ○
3	Syllable: syllable-aware granular	yellow	● ● ○
4	Split: voiced / unvoiced splitter	cyan	○ ○ ●
5	Sequencer: clocked syllable and word steps	blue	● ○ ●

Zone (PITCH)	Algorithm	MEDIA LED	ASSIGN LEDs lit (FEEDBACK · DRY/WET · REVERB)
6 (fully CW)	Chain: voice processing chain	magenta	○ ● ●

The ASSIGN LEDs count the algorithm number in binary: the LED beside FEEDBACK is worth 1, the one beside DRY/WET 2, the one beside REVERB 4.

Notes:

- The zones have a small hysteresis, so the choice does not flicker when PITCH rests near a boundary.
- The press of MEDIA held at power-on does not count as a confirmation: MEDIA must be released and pressed again.
- Holding SEED at power-on still recovers the previous settings, as in the official firmware, and holding FREEZE still starts the bootloader. Do not hold FREEZE together with MEDIA.
- The six single firmwares behave exactly like the corresponding algorithm in vox_multi; they have no selection step.

Common features

Everything in this section works the same way in all six algorithms. The algorithm chapters only describe what differs.

Audio path

The codec runs at 48 kHz. The speech algorithms (PSOLA, LPC, Syllable, Split, Sequencer) analyse and process at 24 kHz internally, which keeps the full bandwidth of speech (about 11 kHz). Chain runs entirely at 48 kHz. The voice is taken from IN L **(7)**; Syllable mixes IN L and IN R. Some algorithms use IN R **(8)** for a second signal (a carrier in LPC, a music bed in Chain), and only when both inputs are patched.

Every algorithm ends with the same output stage: DRY/WET **[J]** crossfades between the input and the processed signal, REVERB **[K]** sets the amount of the stock Beads reverb, and a peak limiter keeps the output below clipping. Two exceptions: in Chain, DRY/WET sets the amount of parallel compression, and in the red sub-mode of Split it crossfades between the split streams and the original.

The speech buffer

PSOLA, LPC, Syllable and Sequencer record the incoming voice into a buffer of 10.9 seconds (24 kHz, 16-bit) and analyse it as it arrives. TIME **[D]** then works in one of two ways:

- **Live** (FREEZE off): the buffer keeps recording, and TIME is counted back from the present. Fully counter-clockwise is the most recent material.
- **Frozen** (FREEZE on): recording stops, and TIME addresses the frozen buffer from its oldest end, so it scrubs through a fixed recording.

The FREEZE button **[A]** toggles the frozen state; the FREEZE gate input **(9)** freezes while it is high. The FREEZE LED shows the state.

Knobs, attenuverters and CV inputs

TIME, SIZE, SHAPE and PITCH each have an attenuverter **[M]** and a CV input **(2)–(5)**. The CV is scaled by the attenuverter and added to the knob: with the attenuverter fully clockwise, +5 V sweeps the whole range of the knob. The attenuverters have a dead zone at noon.

PITCH **[E]** follows the official Beads law in every algorithm except Chain: ±24 semitones with resting points at the musical intervals (1, 3, 4, 7, 12 and 24 semitones). PITCH CV **(5)** follows 1 V/octave, scaled by the PITCH attenuverter. In Chain, PITCH is a plain 0–1 knob.

DENSITY **[C]** has no attenuverter: its CV input **(6)** is added to the knob position directly.

MEDIA: sub-modes and input gain

Each algorithm has four sub-modes, shown by the colour of the MEDIA LED. The sub-modes of each algorithm are listed in its chapter.

Action	Result
Press MEDIA **[F]**	Next sub-mode
Hold MEDIA for 1.5 s	Back to automatic input gain, and re-learn it
Hold MEDIA and turn FEEDBACK **[I]**	Manual input gain; the MEDIA LED becomes a level meter
MEDIA + SEED, MEDIA + FREEZE	Algorithm options (see each chapter)

The automatic input gain is the one of the official firmware. It brings any line or modular level to a useful range, and the VU LED blinks while it adapts.

ASSIGN: the assignable CV input

The ASSIGN CV input **(1)** can modulate FEEDBACK **[I]**, DRY/WET **[J]** or REVERB **[K]**, as in the official firmware. Press ASSIGN **[L]** to step through the destinations; the LED beside the selected knob lights. Some algorithms add a fifth destination after REVERB, shown by all three ASSIGN LEDs blinking together; its function is given in the algorithm's chapter.

To modulate several knobs at once, hold ASSIGN and turn FEEDBACK, DRY/WET and REVERB: each knob then sets the amount of ASSIGN CV it receives, and the three LEDs show the amounts.

Gate output

ASSIGN + SEED steps through the gate output modes. When a gate mode is active, OUT R **(12)** carries a gate or trigger signal derived from the analysis (a voiced gate, onset triggers, step triggers), and OUT L **(11)** carries the mono mix of both channels. Each algorithm's chapter lists its gate modes. Press ASSIGN + SEED until the mode is off to get stereo audio back.

Other gestures

- **ASSIGN + MEDIA** starts the CV calibration. The procedure and the LED sequence are those of the official firmware, and the calibration is shared with it.
- **ASSIGN + FREEZE** toggles the autosave flag of the official firmware. VOX saves its own settings at the moment they are changed, so the flag has no effect in VOX.
- **SEED held at power-on** recovers the previous settings, as in the official firmware.

LEDs

The VU LED next to the inputs shows the input level, and each algorithm adds its own indications (a flash on every detected syllable, a blink on every step). The FREEZE and SEED LEDs show the state of their buttons and of any latch.

1 · PSOLA — pitch-synchronous voice (green)

PSOLA changes the pitch of a voice without changing its timbre, stretches or scrubs speech without changing its pitch, and reshapes its melody: flattened to a monotone, exaggerated, or snapped to a musical scale.

How it works

A pitch tracker follows the voice as it is recorded and places a mark on every glottal pulse (every period of the voice). Playback rebuilds the voice from short grains, each two periods long and centred on a mark, laid down at a new spacing: closer together for a higher pitch, further apart for a lower one. Because each grain keeps the shape of the original period, the formants (the vocal tract resonances that make a voice recognisable) stay where they were.

The tracker covers 50 to 600 Hz, wide enough to follow a Morphagene's varispeed. Where the voice is uncertain (speech over music, for example), PSOLA falls back to 30 ms grains; on unvoiced sounds (s, f, sh) it uses short grains at random offsets, so the result does not buzz.

Controls

Control	Function
TIME **[D]**	Live: how far behind the input the voice is read (at least 100 ms). Frozen: position in the buffer
DENSITY **[C]**	Read speed. Noon: hold, a sustained vowel. Clockwise: 0 to 2×, with a flat zone at 1× around 3 o'clock. Counter-clockwise: reverse, 0 to −2×
PITCH **[E]**	Transposition with the formants kept in place
SIZE **[G]**	Formant shift, ±12 semitones (the size of the vocal tract). Noon: natural. In red: number of repeats per period
SHAPE **[H]**	Intonation. Noon: natural. Clockwise: towards a monotone on the PITCH note (C3 with PITCH at 0). Counter-clockwise: the melody is exaggerated, up to 3× its distance from the speaker's average pitch. In yellow: strength of the scale correction
FEEDBACK **[I]**	Re-records the output into the buffer, so repeated transpositions build up
DRY/WET **[J]**, REVERB **[K]**	Output stage (see Common features)
FREEZE **[A]** / **(9)**	Stops recording and analysis; TIME scrubs the frozen buffer

Control	Function
SEED **[B]** / **(10)**	Live: holds the current period while held, for a stutter or a sustained tone. Frozen: jumps back to the TIME position

Sub-modes (MEDIA)

LED	Sub-mode
Green	PSOLA
Yellow	Scale: the pitch is quantized to a scale whose root is the PITCH setting, with a 20 ms glide between notes
Red	Period stutter: each period is repeated 1 to 8 times (clockwise half of SIZE), for vocal fry and robotic voices
Blue	Narrow range: the tracker is limited to 70–400 Hz, which avoids octave errors on ordinary speech

Options and indicators

- **SEED + FREEZE**: latch. The section between the TIME position and the current read position loops.
- **MEDIA + SEED**: scale for the yellow sub-mode: chromatic, major, minor, pentatonic (0 to 3 ASSIGN LEDs).
- **Gate output** (ASSIGN + SEED): OUT R is high while the material at the read position is voiced.
- **VU LED**: input level. It blinks when the pitch tracker has lost the voice for more than 200 ms.

Patching ideas

- Turn SHAPE fully clockwise and sequence PITCH CV: the speech is sung on the notes of the sequence.
- Set DENSITY to noon and play PITCH from a keyboard: a held vowel becomes a playable voice.
- With FREEZE on, sweep TIME slowly with an LFO for a pitch-stable scrub through a sentence.

2 · LPC — analysis and resynthesis (red)

LPC takes a voice apart into its articulation (the shape of the vocal tract) and its source (the buzz of the vocal folds, or the hiss of breath), and lets you replace, warp, freeze or re-time each one. It whispers, speaks like a robot, sings on a fixed note, turns a synthesizer into a talking instrument, and imitates the speech chips of early-1980s toys.

How it works

Every 10 ms, a 24th-order linear-prediction analysis measures the spectral envelope of the voice at the read position. An inverse filter built from it removes the envelope and leaves the residual, a flat signal that carries the source. A synthesis filter then puts an envelope back onto an excitation chosen with SHAPE. With SHAPE at noon and SIZE at noon, the synthesis filter is the exact inverse of the analysis filter and the voice comes out unchanged, delayed by 62.5 ms.

A pitch tracker records whether each moment of the buffer is voiced and at what pitch, so the pulse excitation can follow the speaker's intonation.

Controls

Control	Function
TIME **[D]**	Live: how far behind the input the voice is read. Frozen: position in the buffer
DENSITY **[C]**	Speed at which the articulation moves. Noon: hold, a frozen vowel shape. Clockwise: 0 to 2×, with a flat zone at 1×. Counter-clockwise: reverse
PITCH **[E]**	Pitch of the pulse excitation. When the pulse follows the voice, PITCH transposes it; when the pulse is fixed, PITCH sets the note (C3 at 0)
SIZE **[G]**	Formant warp, ±6 semitones, with the pitch unchanged. Noon: natural. In red: the chip clock, which moves pitch and formants together
SHAPE **[H]**	Excitation, counter-clockwise to clockwise: noise (whisper) → automatic (pulse on vowels, noise on consonants) → noon: residual (the original voice) → pulse that follows the voice → pulse on a fixed note (robot, monotone)
FEEDBACK **[I]**	Slew: smoothing of the articulation, 0 to 500 ms, for slurred, legato speech

Control	Function
DRY/WET **[J]**, REVERB **[K]**	Output stage
FREEZE **[A]** / **(9)**	Stops recording
SEED **[B]** / **(10)**	Live: holds the current articulation while held. Frozen: jumps back to the TIME position

Sub-modes (MEDIA)

LED	Sub-mode
Green	Resynthesis: SHAPE as described above, with the residual at noon
Red	Speaking chip: a lo-fi vocoder modelled on early speech-synthesis chips (8 kHz, 10th order, 25 ms frames, quantized parameters, 8-bit output). SIZE detunes the chip's clock
Blue	Synth excitation: the same SHAPE range without the residual, from noise through automatic to pulse
Magenta	Cross-synthesis: SHAPE crossfades from the residual to the signal on IN R **(8)**, so the voice articulates the carrier like a talk box. The MEDIA LED blinks when IN R is not patched

The speaking chip is built from original parameters; no data from any commercial chip is used.

Options and indicators

- **SEED + FREEZE**: latch. The current articulation is held indefinitely: a formant drone to play with PITCH CV.
- **MEDIA + SEED**: pitch following. The pulse follows either the detected pitch, or a smoothed version of it (200 ms) that removes jitter. The ASSIGN LEDs show the setting.
- **Gate output** (ASSIGN + SEED): OUT R is high while the voice is voiced.
- **VU LED**: input level; in magenta, the level of the carrier on IN R.

Patching ideas

- Magenta, with a sawtooth or a chord from an oscillator on IN R: the classic talking synthesizer. Bright carriers rich in harmonics articulate best.
- Latch a vowel (SEED + FREEZE) and play PITCH from a sequencer for a choir of one syllable.
- SHAPE fully counter-clockwise and FEEDBACK at 3 o'clock: a slow, slurred whisper.

3 · Syllable — syllable-aware granular (yellow)

Syllable is a granular processor whose grains begin and end on the boundaries of speech: onsets, syllables or whole words. Speech stays intelligible when you want it to, and breaks apart in units of language rather than at arbitrary points when you don't.

How it works

As the voice is recorded, a segmenter finds the onsets (the start of every sound), the syllables (from the rhythm of the vowels) and the words (from the pauses). Each finished unit is added to a table and becomes playable about 50 to 150 ms after it ends. Grains are then taken from that table: TIME chooses which unit, SIZE how much of it, SHAPE how much the choice is scattered. Up to six grains play at once.

Controls

Control	Function
TIME **[D]**	Which unit is played. Live: fully counter-clockwise is the newest finished unit, turning clockwise goes further back. Frozen: fully counter-clockwise is the oldest unit in the buffer. The positions are quantized to units, so a CV sequencer can address words directly
SIZE **[G]**	Grain length. Counter-clockwise: a fraction of the unit, from its start (crumbs). Noon: exactly one unit. Clockwise: up to 8 consecutive units
SHAPE **[H]**	Scatter and envelope. Counter-clockwise: the exact unit with sharp edges. Noon: random choice within ±2 units, softer fades. Clockwise: within ±16 units, with up to 50% of the grains reversed

Control	Function
DENSITY **[C]**	Grain generator. Noon: off. Clockwise: regular, 0.5 to 20 grains per second. Counter-clockwise: random, 0.5 to 20 per second on average. When the SEED input is patched, only it triggers grains
PITCH **[E]**	Transposition (resampling)
FEEDBACK **[I]**	Re-records the output into the buffer, where it is segmented again: fragments of fragments
DRY/WET **[J]**, REVERB **[K]**	Output stage
FREEZE **[A]** / **(9)**	Stops recording; the units keep their addresses, so TIME positions stay put
SEED **[B]** / **(10)**	Plays a grain at the TIME unit. A Morphagene's EOSG output on SEED triggers a grain at every splice

Sub-modes (MEDIA)

LED	Unit
Green	Syllable
Yellow	Word
Red	Onset slice: from one onset to the next, the finest unit
Blue	Syllable, with pause removal

Pause removal: silences longer than 150 ms are not recorded. The buffer then holds speech rather than time, and the 150 ms kept at each pause preserve the word boundaries.

Options and indicators

- **SEED + FREEZE**: latch. The last unit loops seamlessly; the SEED LED blinks.
- **MEDIA + FREEZE**: pause removal on or off, in any sub-mode (all ASSIGN LEDs flash = on, none = off). While it is on outside the blue sub-mode, the MEDIA LED is dimmed.
- **MEDIA + SEED**: segmentation sensitivity: low, medium, high (1, 2 or 3 ASSIGN LEDs flash). A higher setting finds more boundaries, a lower one fewer.

- **Gate output** (ASSIGN + SEED): a 5 ms trigger on OUT R at every onset of the live input, or at every word start in the yellow sub-mode.
- **VU LED**: input level; it flashes each time a unit is finished.

Patching ideas

- Patch the onset triggers (gate output) into the SEED input of another module, or back into Syllable's own SEED input, to play grains in the rhythm of the speech.
- Yellow sub-mode, FREEZE on, and a stepped random voltage into TIME CV: the frozen sentence is re-spoken in a random word order.
- Red sub-mode, SIZE counter-clockwise, DENSITY fully clockwise: a stream of consonant crumbs.

4 · Split — voiced / unvoiced splitter (cyan)

Split hears the difference between the vowels of speech and its consonants, and treats them separately. Vowels can ring on as pitched tails while the consonants echo in a delay, the two can swap roles, or they can simply leave the module on separate outputs for the rest of the system to process.

How it works

A classifier sorts every few milliseconds of speech into four classes: silence; voiced (vowels, nasals, voiced consonants such as l, m, n); unvoiced (fricatives and sibilants such as s, f, sh); and bursts (the release of plosives such as p, t, k). It looks at zero crossings, spectral tilt, the balance of high and low frequencies, level, and the periodicity reported by a pitch tracker.

The audio is delayed by 10 ms so that each class switches on just before the sound it applies to: plosives are never clipped. The whole module is therefore 10 ms late. Voiced fricatives (z, v) are split at 4 kHz, the low band going to the voiced stream and the high band to the unvoiced one. The streams always add up exactly to the input.

Controls

Control	Function
SHAPE **[H]**	Balance of the streams. Noon: everything. Clockwise: vowels only, a mumbled hum with its tails. Counter-clockwise: consonants and bursts only, a whispered skeleton. In blue: clockwise attenuates the consonants (down to −24 dB), counter-clockwise the noise between words (down to −40 dB)
SIZE **[G]**	Length of the vowel tails. Fully counter-clockwise: off. Up to a 4 s decay
PITCH **[E]**	Transposition of the tails, which become harmonised drones
TIME **[D]**	Consonant delay time, 10 ms to 1.2 s. With a clock on SEED: ¼, ⅓, ½, ¾, 1, 1½ or 2 clock periods
FEEDBACK **[I]**	Delay feedback, up to 105%, with a 300 Hz high-pass filter and soft clipping in the loop
DENSITY **[C]**	Classifier bias. Noon: neutral. Clockwise: more material counted as voiced. Counter-clockwise: more counted as unvoiced. A CV here reclassifies the voice rhythmically
DRY/WET **[J]**	Output stage (the dry signal is delayed by 10 ms to stay aligned). In red: split versus original
REVERB **[K]**	Reverb on the stream chosen by the sub-mode: the consonants in green, the vowels in yellow, everything in blue
FREEZE **[A]** / **(9)**	Holds every tail indefinitely: a drone of the last vowels
SEED **[B]** / **(10)**	Captures a tail now, even in the middle of a vowel. A regular clock on the SEED input also sets the delay time

Sub-modes (MEDIA)

LED	Sub-mode
Green	Process: vowels into tails, consonants into the delay, reverb on the consonants
Yellow	Swap: vowels into the delay and the reverb, consonants into noise tails (hiss pads)
Red	Splitter: no processing. OUT L carries the vowels (and silence), OUT R the consonants and bursts
Blue	Clean-up: SHAPE works as a de-esser by class, or as a gate on the noise between words

Options and indicators

- **SEED + FREEZE**: latch "mute input vowels": only the tails and the consonants pass. The SEED LED blinks.
- **MEDIA + SEED**: classifier preset: speech, speech over music (relies more on periodicity), telephone (band-limited audio).
- **Gate output** (ASSIGN + SEED): OUT R is high on voiced sounds, aligned with the audio. Leave it off in the red sub-mode, where OUT R carries the consonants.
- **VU LED**: the classifier at work. Off: silence. Dim: unvoiced. Bright: voiced. Flashing: a burst.

Patching ideas

- Red sub-mode: send the consonants (OUT R) through a resonant filter or a short delay, and mix them back with the vowels (OUT L).
- Green, SIZE at 2 o'clock, PITCH at +7: every vowel leaves a fifth-up drone behind it.
- Clock SEED from the tempo of the music under the speech: the consonant echoes fall in time.

5 · Sequencer — clocked syllables and words (blue)

Sequencer locks speech to a clock. The syllables or words it finds in the voice become the steps of a pattern, played in order, backwards, shuffled or at random, either cut to the length of the step or time-stretched to fill it exactly, with the pitch kept. Stutters and ratchets add repeats.

How it works

The voice is recorded and segmented as in Syllable, and the finished units form a pool. On every clock step, the pattern picks a unit from the pool and plays it on one of two alternating voices. In the stretch sub-modes, a unit is time-stretched or compressed to the step length (WSOLA, which keeps the pitch) when the ratio is between 0.25× and 4×; outside that range it is cut.

Controls

Control	Function
TIME **[D]**	Where the pattern starts in the pool. Live: counted back from the newest unit. Frozen: from the oldest
SIZE **[G]**	Pattern length, 1 to 16 steps
SHAPE **[H]**	Order, in five zones: in order → reverse → fixed shuffle → new random order each cycle → random on every step
DENSITY **[C]**	With a clock on SEED: ÷4, ÷3, ÷2, ×1, ×2, ×3, ×4 (seven zones). Without a clock: tempo, 40 to 240 BPM, in sixteenth-note steps
PITCH **[E]**	Transposition, sampled at the start of each step, so a CV sequencer can play the words as a melody
FEEDBACK **[I]**	Repeat: probability that a step plays the previous unit again (stutter). Above 3/4 of the travel, steps can also ratchet into 2 to 4 repeats
DRY/WET **[J]**, REVERB **[K]**	Output stage
FREEZE **[A]** / **(9)**	Freezes the buffer and the pool: the pattern draws from fixed material
SEED input **(10)**	Clock input. The period is smoothed over 3 pulses; a tempo change of more than 20% is followed after 2 pulses
SEED button **[B]**	Without a clock: tap tempo (quarter notes, average of 3 taps; moving DENSITY returns to the knob tempo). With a clock: advances one step by hand

Sub-modes (MEDIA)

LED	Unit	Steps
Green	Syllable	Cut to the step
Yellow	Word	Cut to the step
Cyan	Syllable	Stretched to the step
Magenta	Word	Stretched to the step

Options and indicators

- **SEED + FREEZE**: pattern lock. A new fixed shuffle is drawn and the order is held, so the random zones of SHAPE stop changing.
- **MEDIA + FREEZE**: pause removal on or off (the MEDIA LED is dimmed when on), as in Syllable.
- **MEDIA + SEED**: swing: off, 16% or 33% on the even steps (0, 1 or 2 ASSIGN LEDs flash).
- **ASSIGN fifth destination: reset.** All three ASSIGN LEDs blink; a rising edge above 1 V on the ASSIGN input restarts the pattern on the next step.
- **Gate output** (ASSIGN + SEED): off → step triggers (5 ms, on every step and every ratchet) → accent gate (high during steps whose unit is among the loudest quarter of the pool).
- **VU LED**: blinks on every step, brighter on accents.

Patching ideas

- Send the same master clock to the Morphagene's CLK input and to SEED, so that the reel and the pattern stay in time. Or patch the Morphagene's EOSG output into SEED to step at the end of every splice.
- Magenta sub-mode, SIZE at 4 steps, SHAPE on fixed shuffle: a four-word phrase re-spoken in time, every word stretched to the beat.
- Feed the accent gate to an envelope or a drum module to underline the stressed words.

6 · Chain — voice processing chain (magenta)

Chain is a broadcast-style channel strip for the voice: noise gate, de-esser, compressor, tone, character and limiter, with a music bed on IN R that ducks under the voice, as in a radio advert. It is the one algorithm that does not transform speech; it makes it sit.

How it works

The voice on IN L **(7)** passes through, in order: the automatic input gain, a gate, a high-pass filter, a de-esser, a compressor (with a parallel uncompressed path), a tilt equalizer, a character stage, the reverb and a limiter. When both inputs are patched, the music on IN R **(8)** is ducked by the processed voice and mixed into both outputs. Everything runs at 48 kHz with no added latency. The knobs work within ranges set by the preset chosen with MEDIA.

Controls

Control	Function
DENSITY **[C]**	Amount of compression (threshold and ratio together, with automatic make-up gain). Fully counter-clockwise: off
TIME **[D]**	Compressor speed. Counter-clockwise: fast and punchy (1 ms attack, 40 ms release). Clockwise: slow and smooth (30 ms, 800 ms). The release slows down by up to 3× during sustained compression
SIZE **[G]**	Tone tilt. Noon: flat. Counter-clockwise: body (+6 dB at 150 Hz, presence reduced). Clockwise: presence (+5 dB at 3 kHz) and air (+6 dB at 10 kHz)
SHAPE **[H]**	Character. Noon: clean. Clockwise: radio → telephone (300–3400 Hz, 8 kHz μ-law) → megaphone. Counter-clockwise: lo-fi, down to 4 kHz and 6 bits
PITCH **[E]**	De-esser depth (a plain knob in this algorithm). Fully counter-clockwise: off
FEEDBACK **[I]**	Gate. Fully counter-clockwise: off. Clockwise raises the threshold (−70 to −25 dBFS) and the ratio, up to a hard gate
DRY/WET **[J]**	Parallel compression. Counter-clockwise: uncompressed (still gated, filtered and de-essed). Clockwise: fully compressed
REVERB **[K]**	Room ambience
FREEZE button **[A]**	Bypass, with a 10 ms click-free crossfade. The FREEZE LED is lit when bypassed

Control	Function
FREEZE input **(9)**	Mutes the voice while high (20 ms fades)
SEED button **[B]**	Chooses what the VU LED shows: input level → gain reduction → output level → music bed level. The ASSIGN LEDs flash 1, 2, 3 or all to show which
SEED input **(10)**	Push: +6 dB into the compressor's detector while high, to squash accented words. The SEED LED lights while pushing

Presets (MEDIA)

LED	Preset	High-pass	Compression range	Bed ducking
Green	Podcast	80 Hz	−6 to −30 dB threshold, up to 4:1	−12 dB
Yellow	Broadcast	100 Hz	−8 to −38 dB, up to 6:1	−15 dB
Red	Announcer	100 Hz	−10 to −44 dB, up to 10:1	−18 dB
Blue	Raw	150 Hz	−6 to −20 dB, up to 2.5:1	−12 dB

Podcast uses slightly slower time constants; Broadcast adds a little presence and de-essing; Announcer is faster and 1 dB louder; in Raw, SHAPE is twice as sensitive.

Options and indicators

- **SEED + FREEZE**: input gain automatic (as the official firmware) or fixed at unity. The ASSIGN LEDs flash all for automatic, none for unity.
- **ASSIGN fifth destination: ducking depth.** All three ASSIGN LEDs blink; the ASSIGN CV moves the ducking depth by up to ±24 dB around the preset value at ±5 V.
- **Gate output** (ASSIGN + SEED): OUT R is high while the voice is speaking (voice activity); OUT L carries the mix.

Patching ideas

- Put Chain last in a voice patch to bring Morphagene material to a steady, broadcast-ready level.
- Patch the voice activity gate to a VCA on another sound, so that it moves only while someone speaks.
- Turn SHAPE to telephone and the music bed down: an instant phone call over a soundtrack.

Specifications, settings and credits

VOX runs on unmodified Beads hardware (STM32H743 at 400 MHz, PCM3060 codec at 48 kHz) and processes audio in blocks of 12 samples (0.25 ms).

Algorithm	Code size	Latency of the processed signal	Memory used
PSOLA	66 KB	from 100 ms (TIME sets the delay)	speech buffer, 8,192 pitch marks
LPC	83 KB	62.5 ms (resynthesis)	speech buffer, voicing and pitch track
Syllable	65 KB	a unit is playable 50–150 ms after it ends	speech buffer, unit tables
Split	68 KB	10 ms	1.2 s delay line
Sequencer	73 KB	quantized to the clock	speech buffer, unit tables
Chain	67 KB	none	—
vox_multi (all six)	168 KB	as above	shared between the algorithms

- **Speech buffer**: 10.9 s at 24 kHz, 16-bit (512 KB of the D1 memory).
- **Processing load**: measured on a computer, the heaviest algorithm (LPC) takes about 27,000 instructions per audio block, against a budget of about 100,000 processor cycles per block on the module. The load on the module itself has not been measured yet.

Status

All six algorithms are tested on a computer with synthetic speech: 150 automated checks covering segmentation, pitch tracking, transposition, formant warping, timing and stability. First tests on a module (October 2026): PSOLA and LPC sound as intended; Syllable sounds good but was seen to fall silent after a while, which is being looked into. Split, Sequencer, Chain and the vox_multi selection have not been tried on hardware yet.

Settings storage

VOX uses the settings sector of the official firmware without changing its layout. The calibration is shared and never rewritten by VOX. vox_multi keeps the number of the active algorithm, and the sub-mode and options of each of the six algorithms, in the 14 bytes that the official firmware leaves unused. Settings are saved as soon as they change.

Credits and licence

VOX is built on the Beads firmware platform and the stmlib library by Emilie Gillet (Mutable Instruments), released under the MIT licence; the Beads reverb, drivers, audio bootloader and calibration procedure are hers. The VOX code is released under the MIT licence (`vox/LICENSE`). Mutable Instruments is not involved in VOX and does not support it: to go back to the official firmware, see Installation and firmware update.